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- (54) **SYNTHETIC VIEW FOR TRY-ON EXPERIENCE**
- (71) Applicant: **Snap Inc.**, Santa Monica, CA (US)
- (72) Inventors: **Matan Zohar**, Rishon LeZion (IL); **Omri Berg**, Tel Aviv (IL); **Itamar Berger**, Hod Hasharon (IL); **Gal Sasson**, Kibbutz Ayyelet Hashahar (IL)
- (73) Assignee: **SNAP INC.**, Santa Monica, CA (US)
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Primary Examiner — Nurun Flora
(74) *Attorney, Agent, or Firm* — SCHWEGMAN
LUNDBERG & WOESSNER, P.A.

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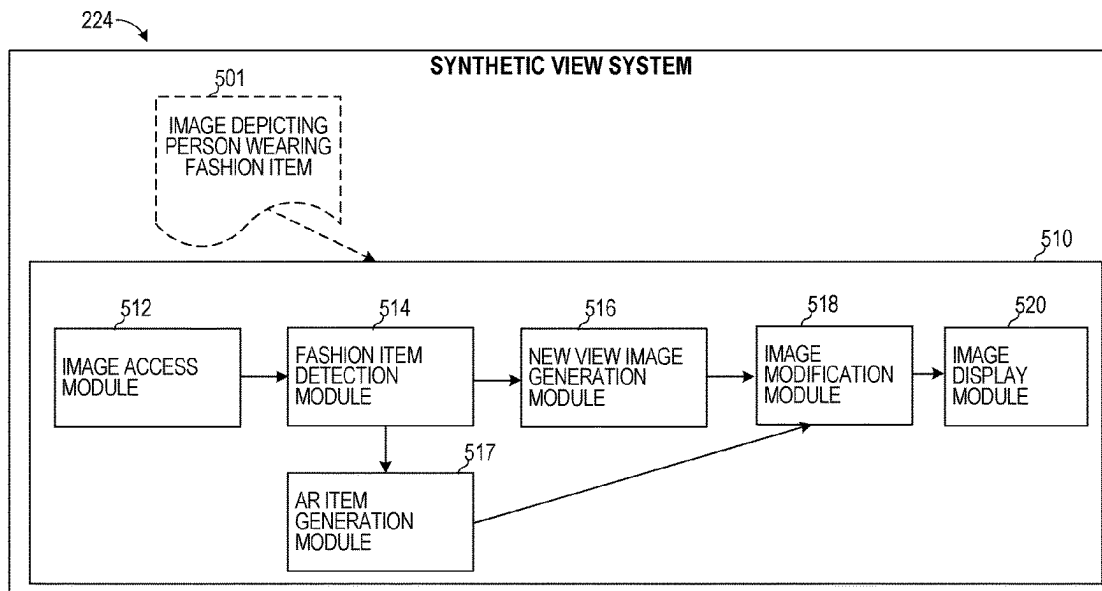
- (57) **ABSTRACT**

Aspects of the present disclosure involve a system for synthetically generating a view of an object to present one or more AR fashion items in an image. The system accesses a first image depicting a first portion of a body of a person from a first camera view. The system detects a fashion item in the first image that is being worn by the person in the first image, the fashion item being depicted in the first image from the first camera view. The system generates an augmented reality (AR) item that visually resembles the fashion item from the second camera view and modifies the second image with the AR item to present a synthetic view of the second portion of the body of the person wearing the AR item corresponding to the fashion item.

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20 Claims, 12 Drawing Sheets



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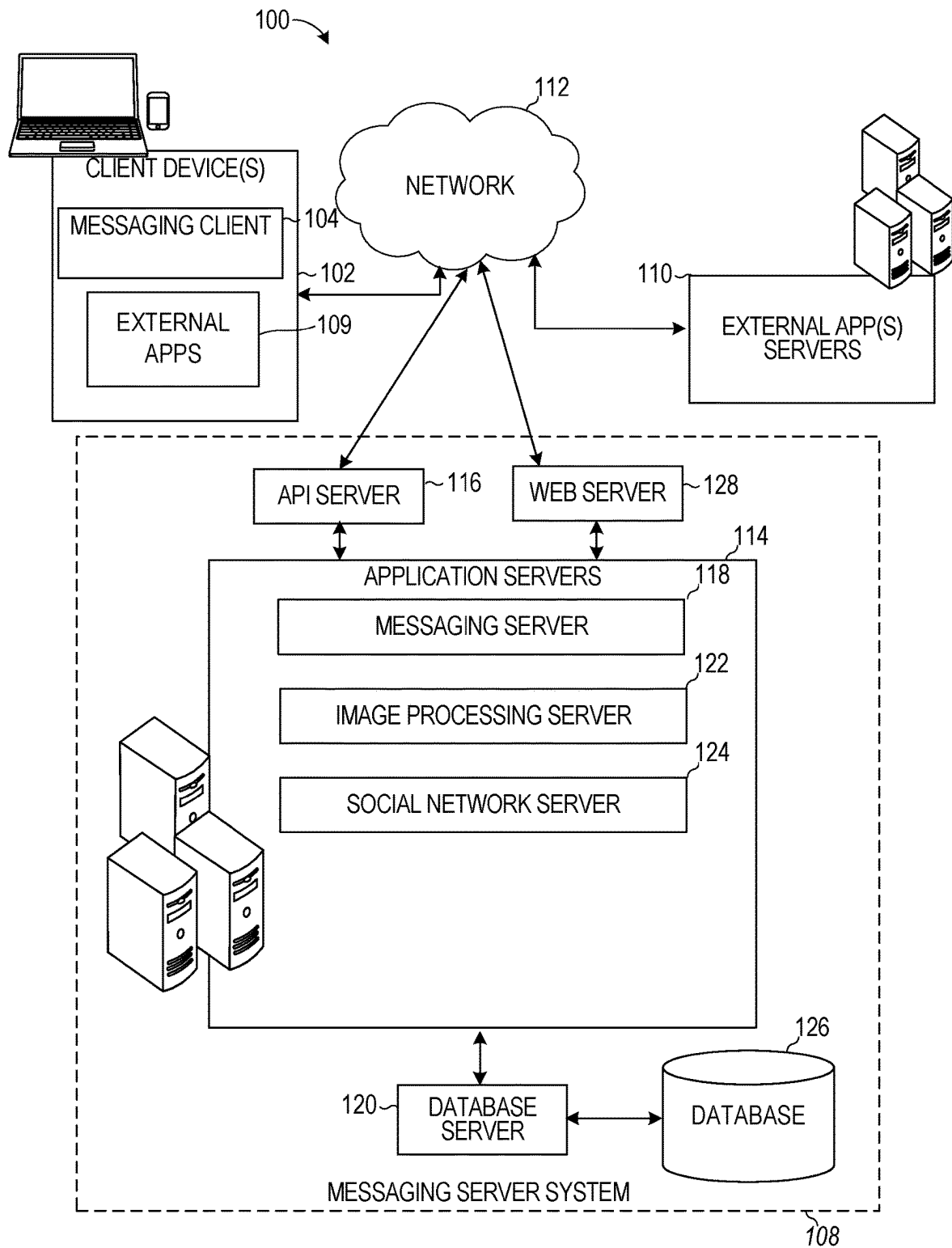


FIG. 1

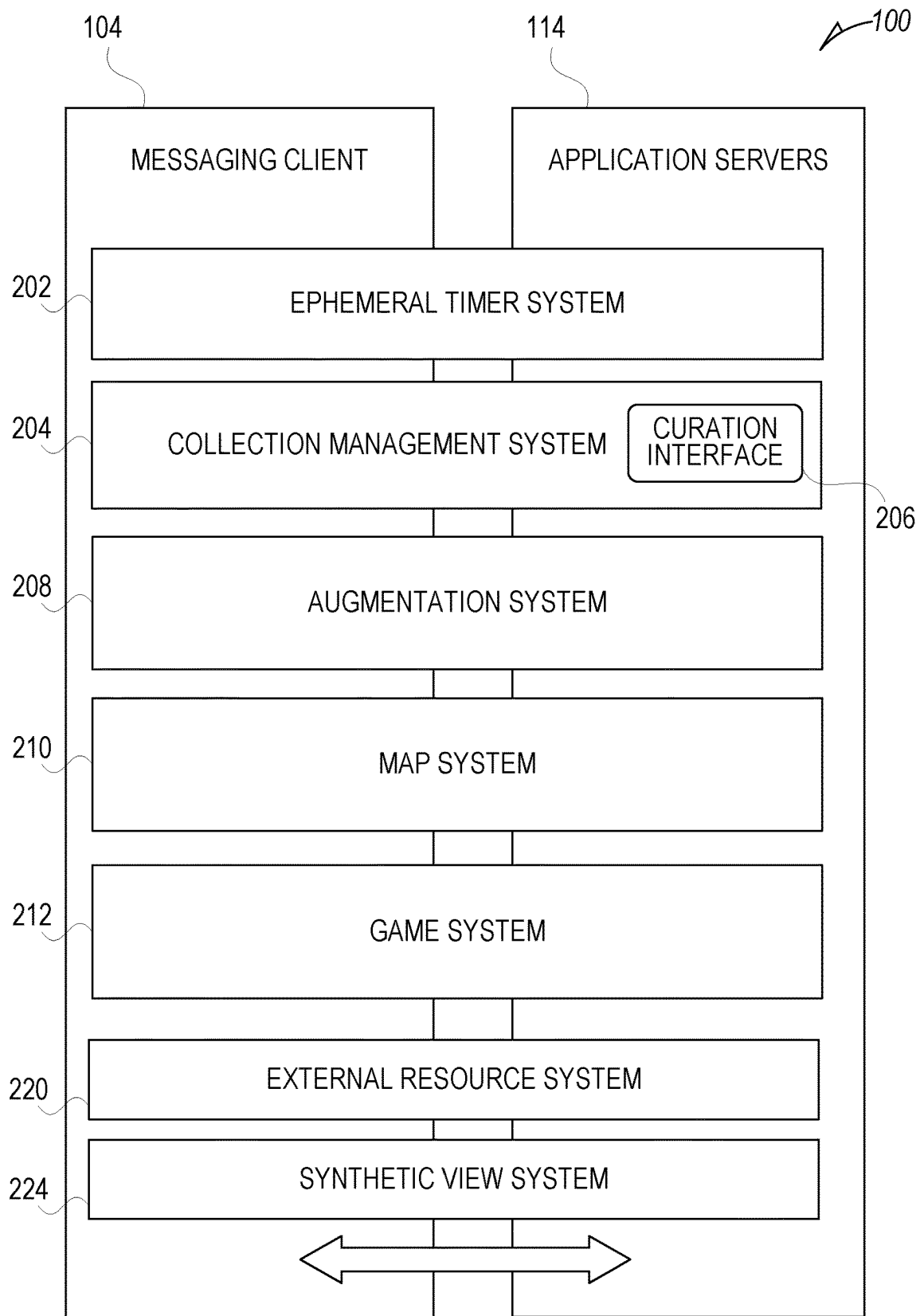


FIG. 2

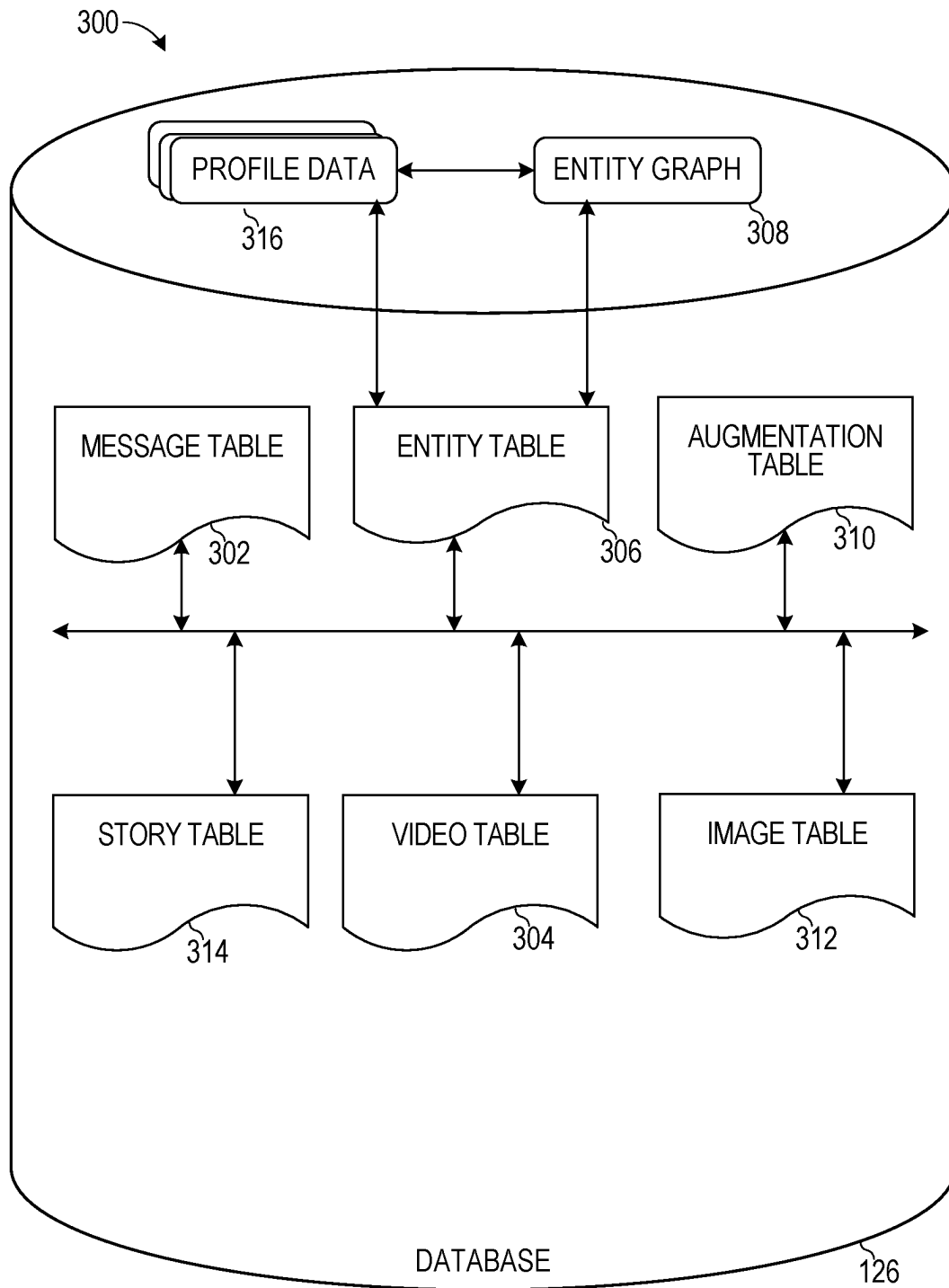


FIG. 3

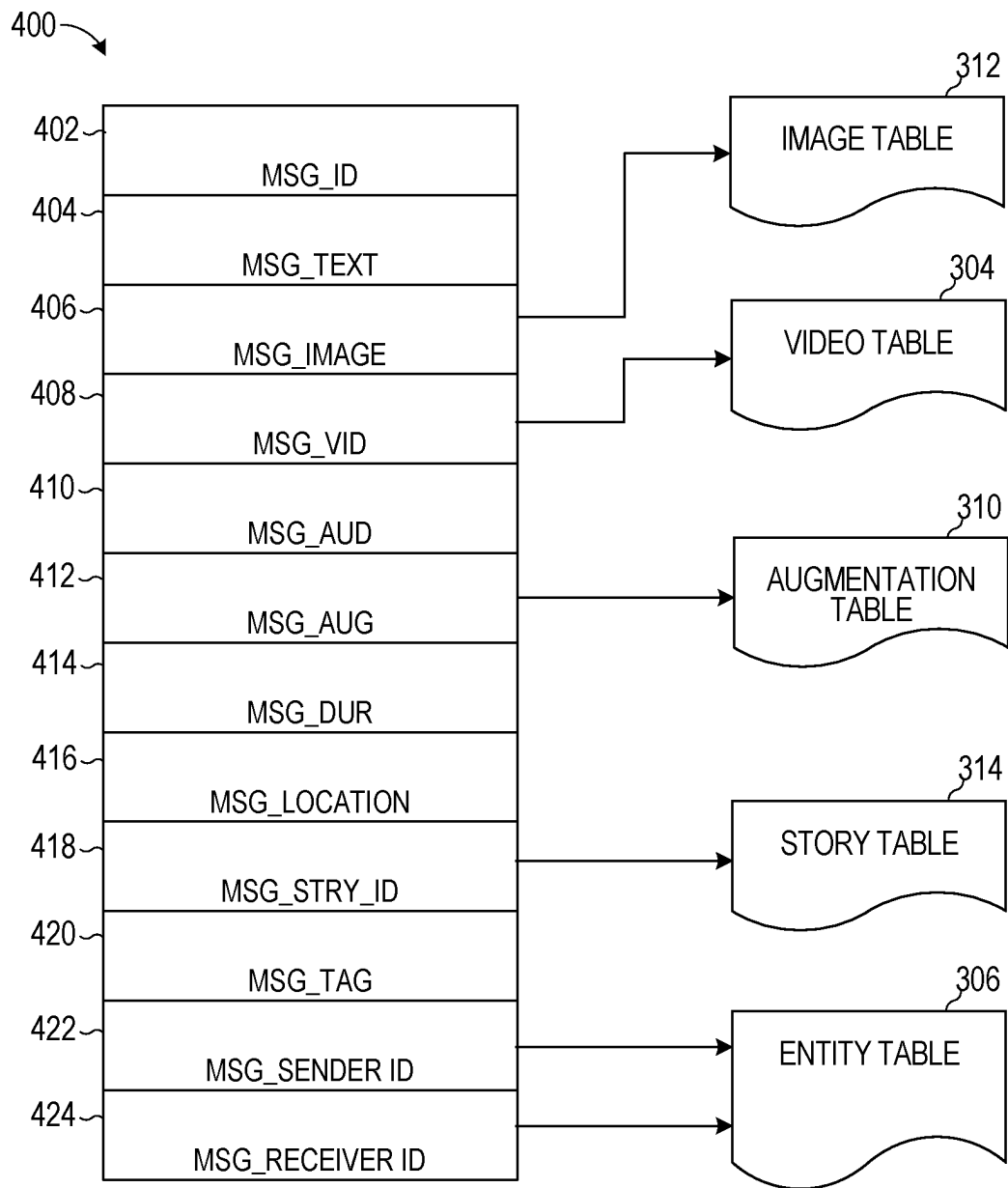


FIG. 4

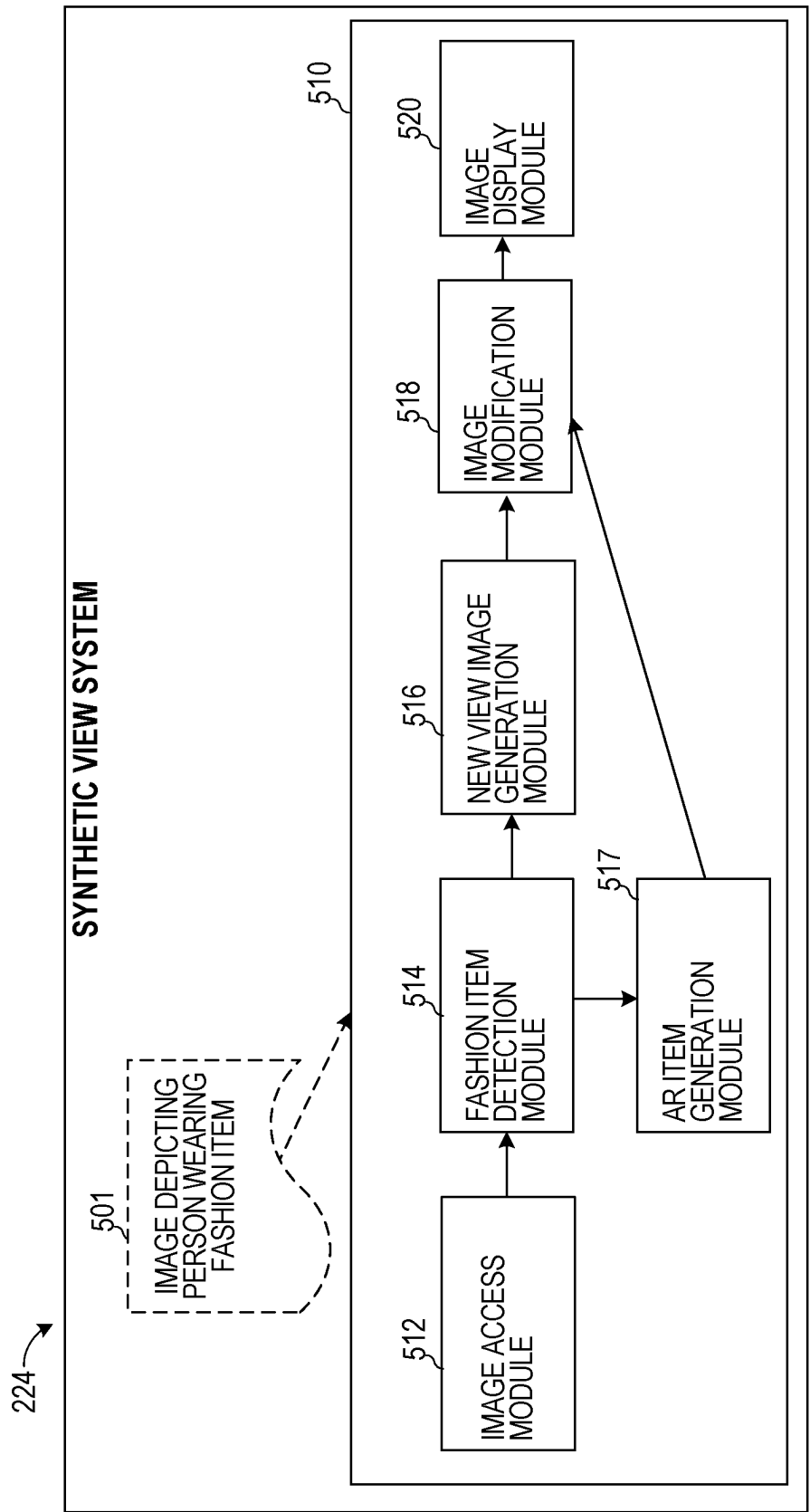
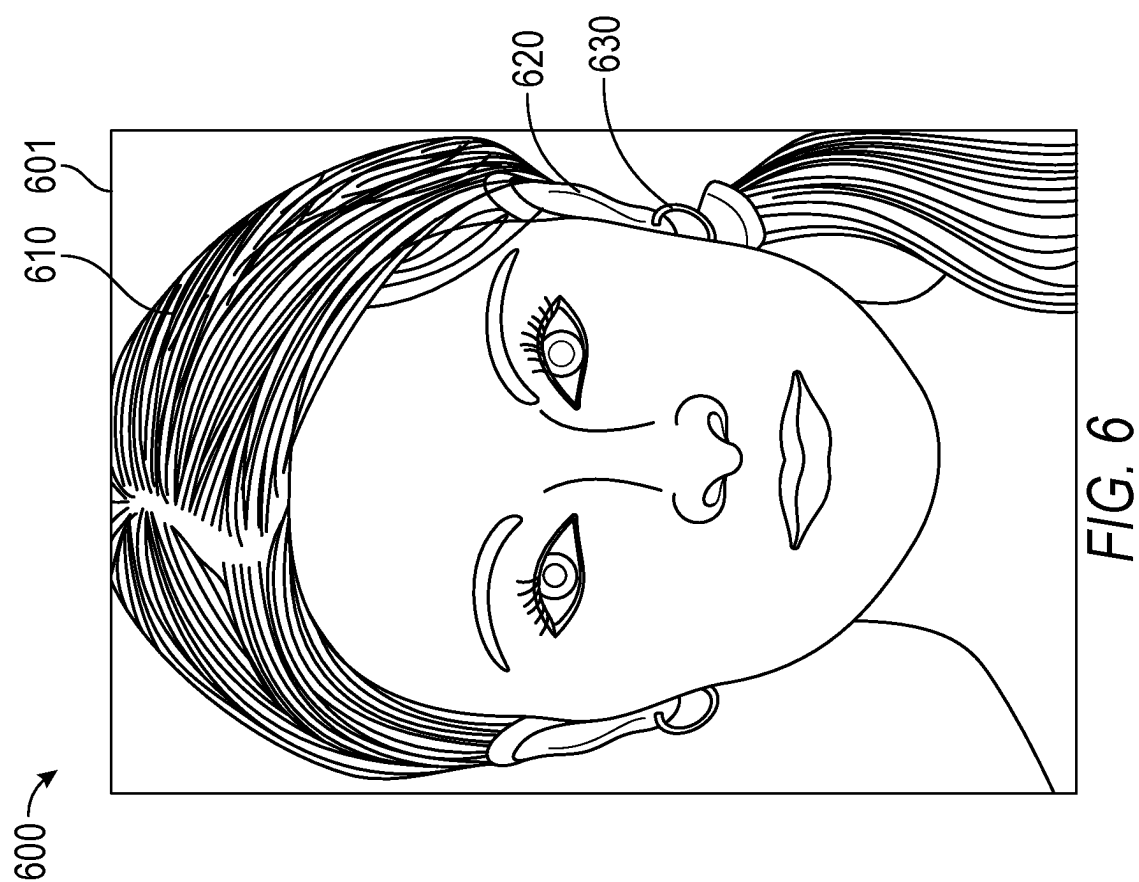
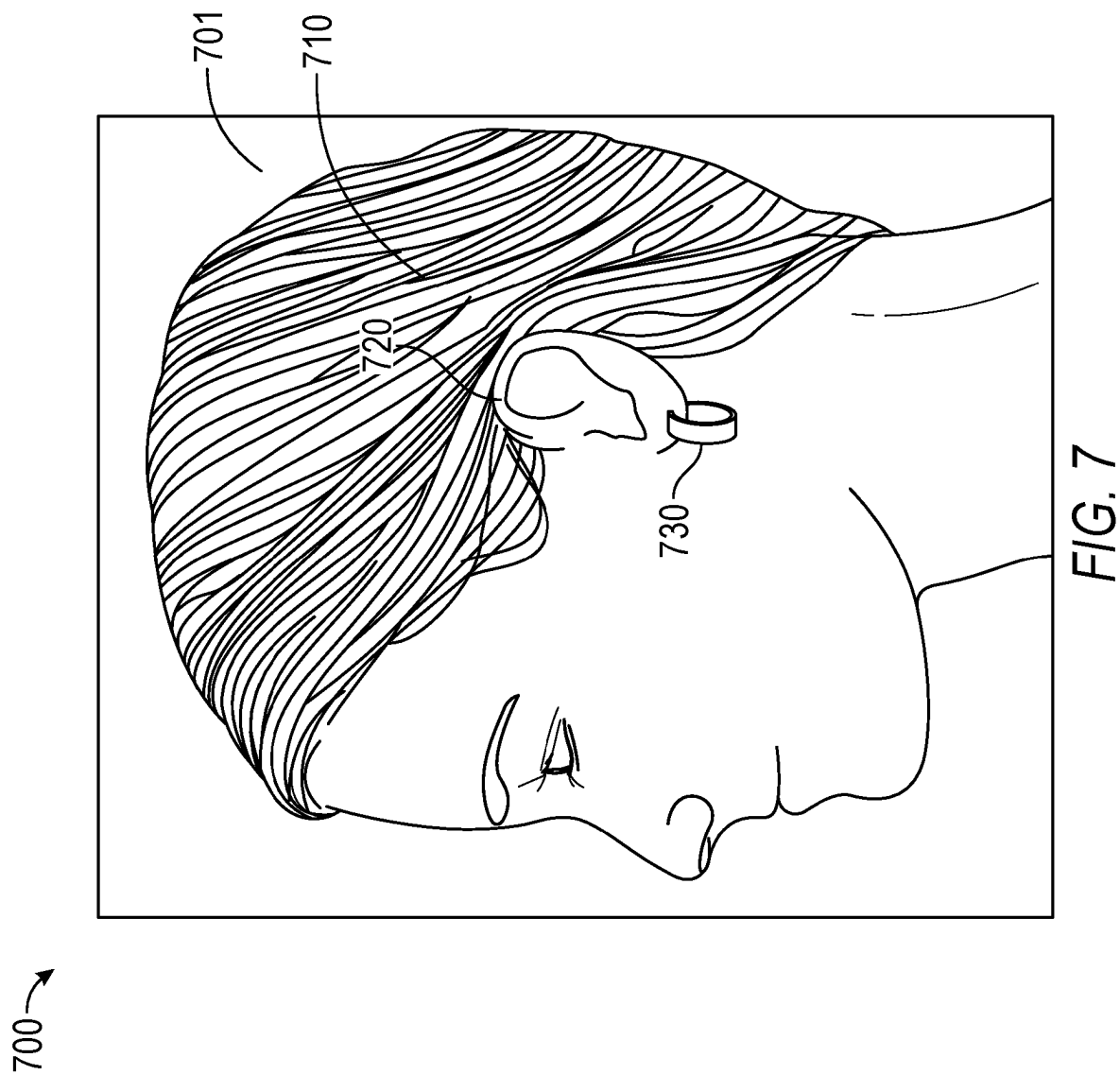


FIG. 5





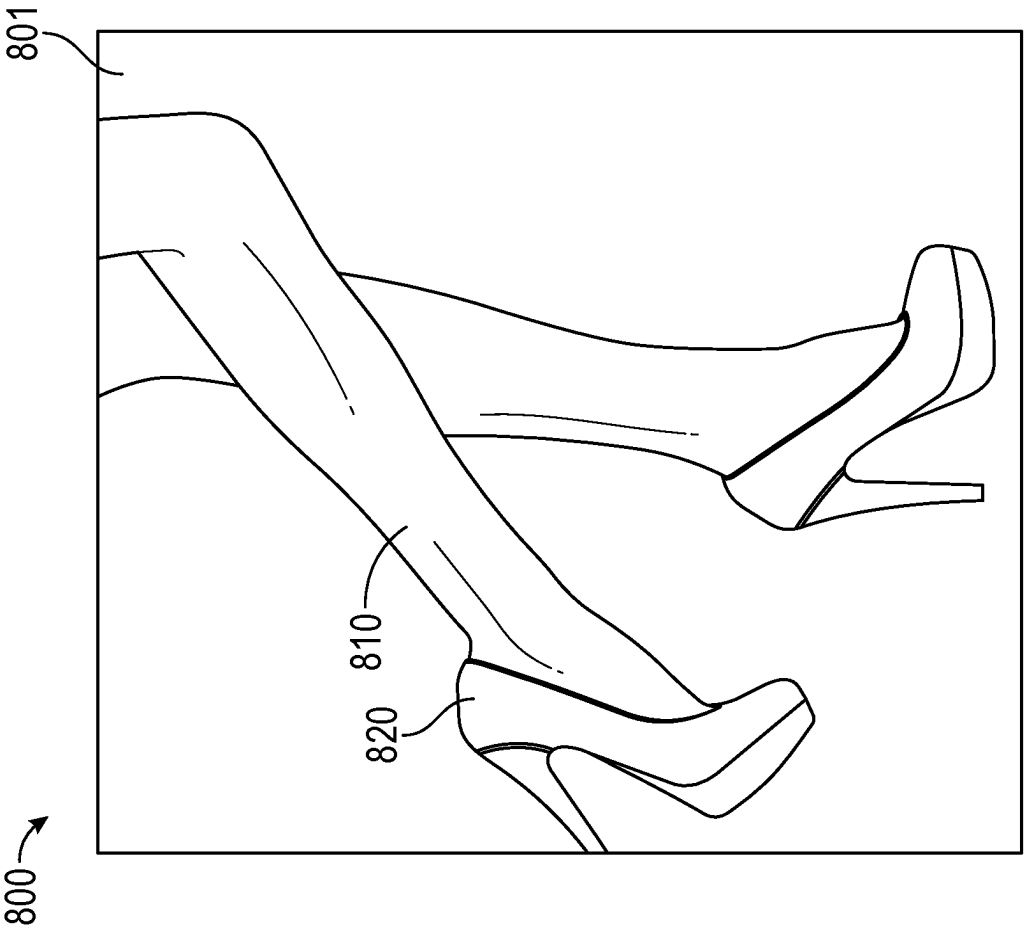
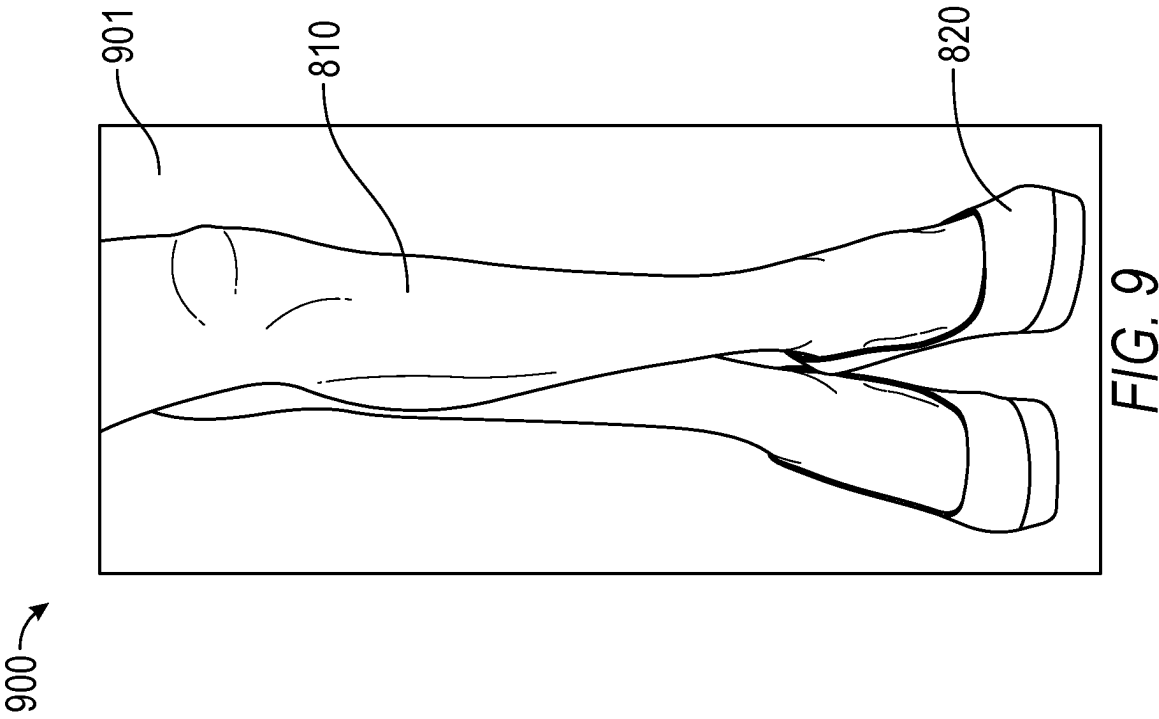


FIG. 8



1000

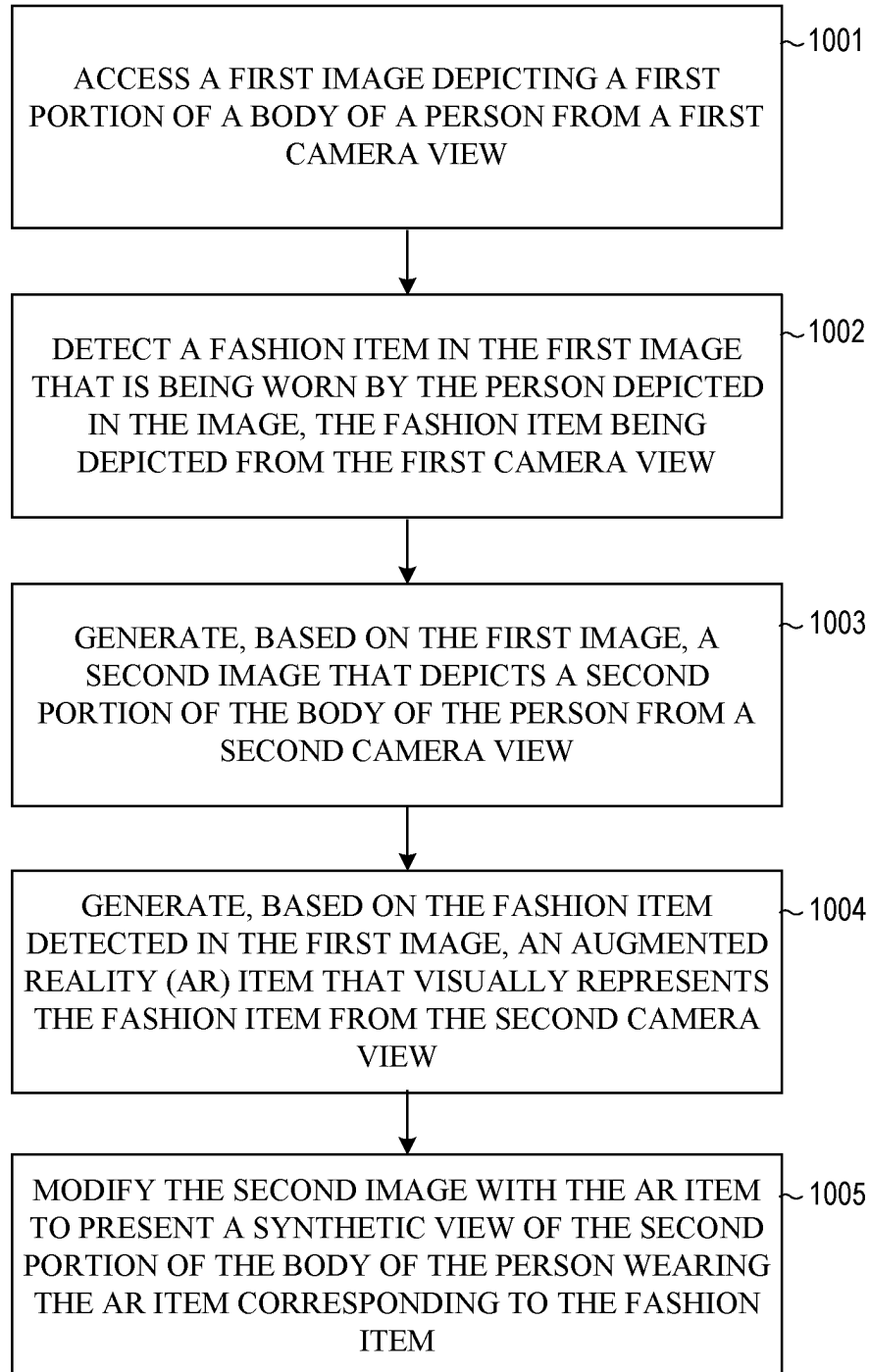


FIG. 10

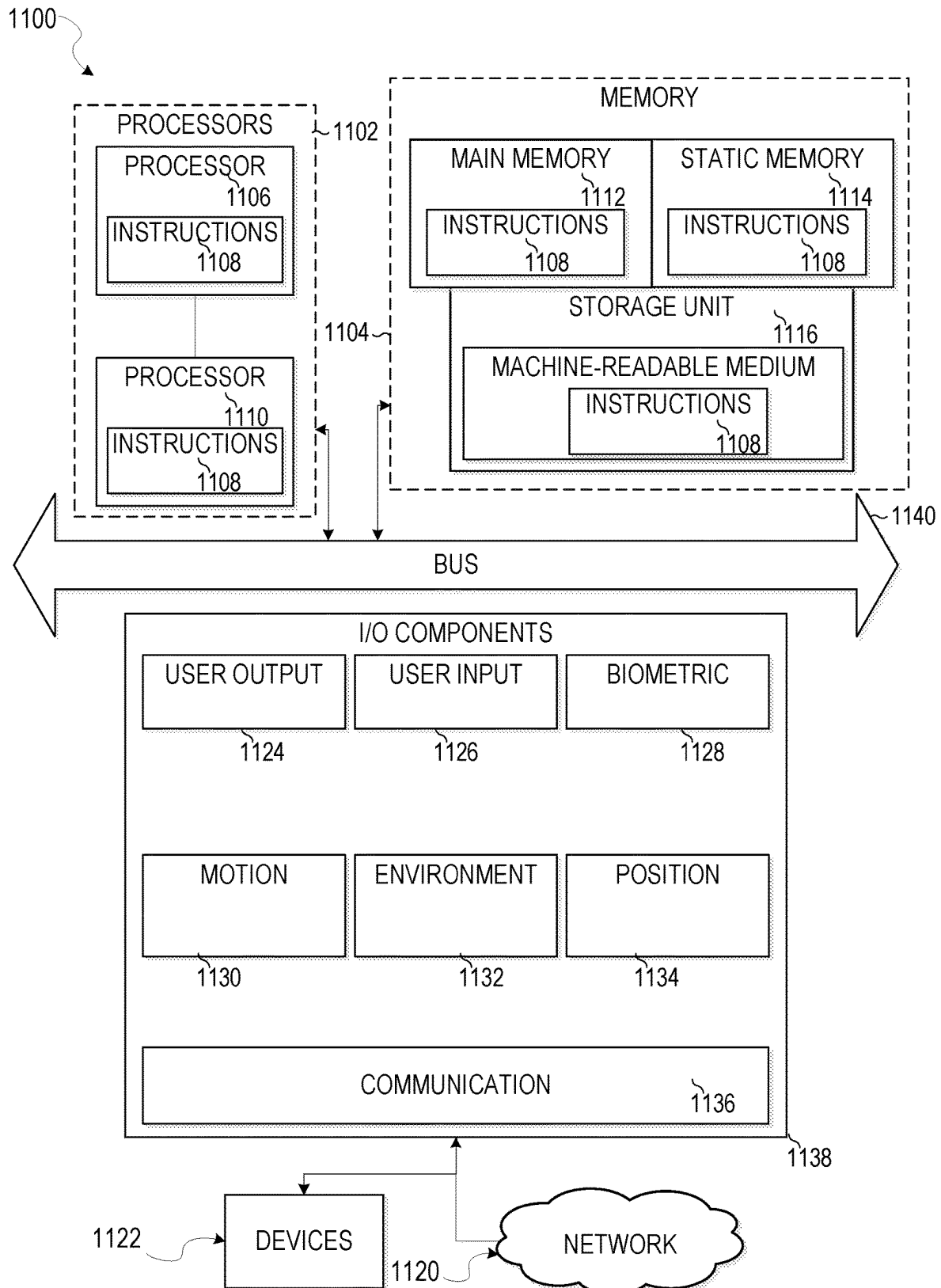


FIG. 11

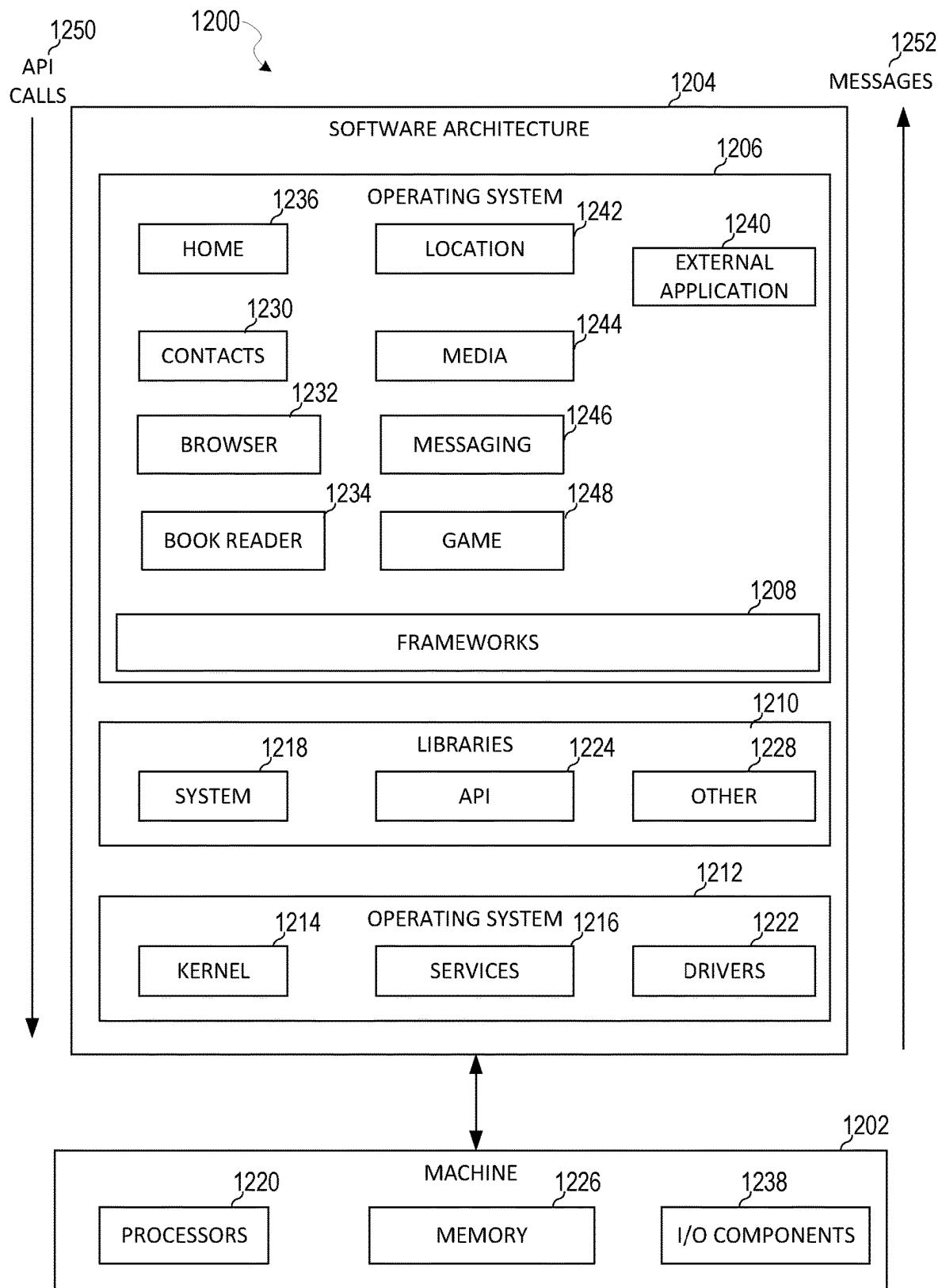


FIG. 12

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SYNTHETIC VIEW FOR TRY-ON EXPERIENCE

TECHNICAL FIELD

The present disclosure relates generally to generating images and providing virtual try-on experiences using a messaging application.

BACKGROUND

Augmented reality (AR) is a modification of a real environment with virtual objects, visual effects or additional information. For example, in virtual reality (VR), a user is completely immersed in a virtual world, whereas in AR, the user is immersed in a world where virtual objects are combined or superimposed on the real world. An AR system aims to generate and present virtual objects that interact realistically with a real-world environment and with each other. Examples of AR applications can include single or multiple player video games, instant messaging systems, and the like.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some nonlimiting examples are illustrated in the figures of the accompanying drawings in which:

FIG. 1 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

FIG. 2 is a diagrammatic representation of a messaging client application, in accordance with some examples.

FIG. 3 is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

FIG. 4 is a diagrammatic representation of a message, in accordance with some examples.

FIG. 5 is a block diagram showing an example synthetic view system, according to some examples.

FIGS. 6-9 are diagrammatic representations of inputs and outputs of the synthetic view system, in accordance with some examples.

FIG. 10 is a flowchart illustrating example operations of the synthetic view system, according to some examples.

FIG. 11 is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, in accordance with some examples.

FIG. 12 is a block diagram showing a software architecture within which examples may be implemented.

DETAILED DESCRIPTION

The description that follows includes systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative examples of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various examples. It will be

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evident, however, to those skilled in the art, that examples may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

Typically, messaging applications and other social network platforms allow users to view a live or real-time camera feed that depicts the users wearing different items. For example, a user can activate a virtual try-on experience in which one or more AR items are placed on the user depicted in the image in real-time. This provides a user with the ability to visualize how different products look on the user before the user purchases the products. Sometimes the items being tried on and depicted as being worn by the users are small and hard to see in the image that is captured by the camera of the device used to present the try-on experience. To get a better idea of how the object looks on the user, the user needs to bring the camera closer to the object of interest or turn/move their bodies or head in awkward ways to get a different perspective.

For example, a user may be wearing earrings and may be depicted as wearing the earrings in a camera feed captured by a front-facing camera of the device. The camera feed may depict the entire face of the user and the user's ears on which the earrings are being worn. However, because the earrings appear relatively small in the overall image, they are hard to see. The user has to manipulate the phone in awkward ways to bring the phone close to the ear of interest on which the earrings are being worn or to obtain a better view, such as to capture an image from a side perspective. This causes a distorted image of the ear and earrings to be presented, and the face of the user may no longer be visible. This makes it difficult for the user to understand and visualize accurately how the earrings look relative to the overall face of the user. Also, maneuvering the camera and the device with the camera in awkward ways to get the right picture of the earrings takes time and is very distracting from the overall experience. Sometimes, facial tracking information used to generate a display of the earrings (in case of AR earrings) is entirely lost when the camera is brought too close to the AR earrings and no longer captures the image of the user's face. When this happens, the AR earrings begin to behave erratically or can be entirely removed from the displayed image, which takes away from the overall illusion that the earrings are part of the real-world image captured by the camera.

The disclosed techniques seek to improve the efficiency of using an electronic device by intelligently automatically generating images of fashion items that are detected in a real-time camera feed from different synthetic views in a simple, seamless, and intuitive manner. The disclosed techniques detect a fashion item being worn by a person depicted in a camera feed that is captured and accessed in real-time. The camera feed may depict a first camera view of the person, such as a front view (showing only the front profile of the face). The disclosed techniques can synthesize a second camera view of the person, such as to depict a side view or side profile of the face using the first camera view image. The disclosed techniques also modify the fashion item to be depicted from the angle corresponding to the first camera view and synthesize a new image that depicts the person wearing the fashion item from the second camera view. This allows the person to capture a single image representing one view of the person wearing a fashion item and seamlessly (without awkwardly maneuvering the camera) access or view one or more additional images that depict the person wearing the same fashion item from different views or perspectives. The additional views can be presented together or separately from the single image that is captured.

This can reduce the overall time and expense incurred by users trying on different fashion items that are real-world fashion items or AR fashion items, such as shoes, shirts, earrings, watches or other fashion items. As used herein, “article of clothing,” “fashion item,” and “garment” are used interchangeably and should be understood to have the same meaning. Article of clothing, garment, or fashion item can include a shirt, pants, skirt, dress, shoe, purse, furniture item, household item, eyewear, eyeglasses, AR logos, AR emblems, pants, shorts, skirts, jackets, t-shirts, blouses, glasses, jewelry, earrings, bunny ears, a hat, earmuffs, or any other suitable item or object.

Specifically, the disclosed techniques access a first image depicting a first portion of a body of a person from a first camera view. The disclosed techniques detect a fashion item in the first image that is being worn by the person and generates, based on the first image, a second image that depicts a second portion of the body of the person from a second camera view. The disclosed techniques generate an AR item that visually resembles the fashion item from the second camera view. The disclosed techniques modify the second image with the AR item to present a synthetic view of the second portion of the body of the person wearing the AR item corresponding to the fashion item. This improves the overall experience of the user in using the electronic device and reduces the overall amount of system resources needed to accomplish a task.

Networked Computing Environment

FIG. 1 is a block diagram showing an example messaging system 100 for exchanging data (e.g., messages and associated content) over a network. The messaging system 100 includes multiple instances of a client device 102, each of which hosts a number of applications, including a messaging client 104 and other external applications 109 (e.g., third-party applications). Each messaging client 104 is communicatively coupled to other instances of the messaging client 104 (e.g., hosted on respective other client devices 102), a messaging server system 108 and external app(s) servers 110 via a network 112 (e.g., the Internet). A messaging client 104 can also communicate with locally-hosted third-party applications (also referred to as “external applications” and “external apps”) 109 using Application Program Interfaces (APIs).

The client device 102 may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the client device 102 may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The client device 102 may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the disclosed operations. Further, while only a single client device 102 is illustrated, the term “client device” shall also be taken to include a collection of machines that individually or jointly execute the disclosed operations.

In some examples, the client device 102 can include AR glasses or an AR headset in which virtual content is displayed within lenses of the glasses while a user views a real-world environment through the lenses. For example, an image can be presented on a transparent display that allows

a user to simultaneously view content presented on the display and real-world objects.

A messaging client 104 is able to communicate and exchange data with other messaging clients 104 and with the messaging server system 108 via the network 112. The data exchanged between messaging clients 104, and between a messaging client 104 and the messaging server system 108, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

The messaging server system 108 provides server-side functionality via the network 112 to a particular messaging client 104. While certain functions of the messaging system 100 are described herein as being performed by either a messaging client 104 or by the messaging server system 108, the location of certain functionality either within the messaging client 104 or the messaging server system 108 may be a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system 108 but to later migrate this technology and functionality to the messaging client 104 where a client device 102 has sufficient processing capacity.

The messaging server system 108 supports various services and operations that are provided to the messaging client 104. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client 104. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information, as examples. Data exchanges within the messaging system 100 are invoked and controlled through functions available via user interfaces of the messaging client 104.

Turning now specifically to the messaging server system 108, an API server 116 is coupled to, and provides a programmatic interface to, application servers 114. The application servers 114 are communicatively coupled to a database server 120, which facilitates access to a database 126 that stores data associated with messages processed by the application servers 114. Similarly, a web server 128 is coupled to the application servers 114, and provides web-based interfaces to the application servers 114. To this end, the web server 128 processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

The API server 116 receives and transmits message data (e.g., commands and message payloads) between the client device 102 and the application servers 114. Specifically, the API server 116 provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client 104 in order to invoke functionality of the application servers 114. The API server 116 exposes various functions supported by the application servers 114, including account registration; login functionality; the sending of messages, via the application servers 114, from a particular messaging client 104 to another messaging client 104; the sending of media files (e.g., images or video) from a messaging client 104 to a messaging server 118, and for possible access by another messaging client 104; the settings of a collection of media data (e.g., story); the retrieval of a list of friends of a user of a client device 102; the retrieval of such collections; the retrieval of messages and content; the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a

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social graph); the location of friends within a social graph; and opening an application event (e.g., relating to the messaging client 104).

The application servers 114 host a number of server applications and subsystems, including, for example, a messaging server 118, an image processing server 122, and a social network server 124. The messaging server 118 implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client 104. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available to the messaging client 104. Other processor- and memory-intensive processing of data may also be performed server-side by the messaging server 118, in view of the hardware requirements for such processing.

The application servers 114 also include an image processing server 122 that is dedicated to performing various image processing operations, typically with respect to images or video within the payload of a message sent from or received at the messaging server 118.

Image processing server 122 is used to implement scan functionality of the augmentation system 208 (shown in FIG. 2). Scan functionality includes activating and providing one or more AR experiences on a client device 102 when an image is captured by the client device 102. Specifically, the messaging client 104 on the client device 102 can be used to activate a camera. The camera displays one or more real-time images or a video to a user along with one or more icons or identifiers of one or more AR experiences. The user can select a given one of the identifiers to launch the corresponding AR experience or perform a desired image modification.

The social network server 124 supports various social networking functions and services and makes these functions and services available to the messaging server 118. To this end, the social network server 124 maintains and accesses an entity graph 308 (as shown in FIG. 3) within the database 126. Examples of functions and services supported by the social network server 124 include the identification of other users of the messaging system 100 with which a particular user has relationships or is “following,” and also the identification of other entities and interests of a particular user.

Returning to the messaging client 104, features and functions of an external resource (e.g., a third-party application 109 or applet) are made available to a user via an interface of the messaging client 104. The messaging client 104 receives a user selection of an option to launch or access features of an external resource (e.g., a third-party resource), such as external apps 109. The external resource may be a third-party application (external apps 109) installed on the client device 102 (e.g., a “native app”), or a small-scale version of the third-party application (e.g., an “applet”) that is hosted on the client device 102 or remote of the client device 102 (e.g., on external resource or app(s) servers 110). The small-scale version of the third-party application includes a subset of features and functions of the third-party application (e.g., the full-scale, native version of the third-party standalone application) and is implemented using a markup-language document. In one example, the small-scale version of the third-party application (e.g., an “applet”) is a web-based, markup-language version of the third-party application and is embedded in the messaging client 104. In

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addition to using markup-language documents (e.g., a *.ml file), an applet may incorporate a scripting language (e.g., a *.js file or a .json file) and a style sheet (e.g., a *.ss file).

In response to receiving a user selection of the option to launch or access features of the external resource (e.g., external app 109), the messaging client 104 determines whether the selected external resource is a web-based external resource or a locally-installed external application. In some cases, external applications 109 that are locally installed on the client device 102 can be launched independently of and separately from the messaging client 104, such as by selecting an icon, corresponding to the external application 109, on a home screen of the client device 102. Small-scale versions of such external applications can be launched or accessed via the messaging client 104 and, in some examples, no or limited portions of the small-scale external application can be accessed outside of the messaging client 104. The small-scale external application can be launched by the messaging client 104 receiving, from an external app(s) server 110, a markup-language document associated with the small-scale external application and processing such a document.

In response to determining that the external resource is a locally-installed external application 109, the messaging client 104 instructs the client device 102 to launch the external application 109 by executing locally-stored code corresponding to the external application 109. In response to determining that the external resource is a web-based resource, the messaging client 104 communicates with the external app(s) servers 110 to obtain a markup-language document corresponding to the selected resource. The messaging client 104 then processes the obtained markup-language document to present the web-based external resource within a user interface of the messaging client 104.

The messaging client 104 can notify a user of the client device 102, or other users related to such a user (e.g., “friends”), of activity taking place in one or more external resources. For example, the messaging client 104 can provide participants in a conversation (e.g., a chat session) in the messaging client 104 with notifications relating to the current or recent use of an external resource by one or more members of a group of users. One or more users can be invited to join in an active external resource or to launch a recently-used but currently inactive (in the group of friends) external resource. The external resource can provide participants in a conversation, each using a respective messaging client 104, with the ability to share an item, status, state, or location in an external resource with one or more members of a group of users into a chat session. The shared item may be an interactive chat card with which members of the chat can interact, for example, to launch the corresponding external resource, view specific information within the external resource, or take the member of the chat to a specific location or state within the external resource. Within a given external resource, response messages can be sent to users on the messaging client 104. The external resource can selectively include different media items in the responses, based on a current context of the external resource.

The messaging client 104 can present a list of the available external resources (e.g., third-party or external applications 109 or applets) to a user to launch or access a given external resource. This list can be presented in a context-sensitive menu. For example, the icons representing different ones of the external applications 109 (or applets) can vary based on how the menu is launched by the user (e.g., from a conversation interface or from a non-conversation interface).

System Architecture

FIG. 2 is a block diagram illustrating further details regarding the messaging system 100, according to some examples. Specifically, the messaging system 100 is shown to comprise the messaging client 104 and the application servers 114. The messaging system 100 embodies a number of subsystems, which are supported on the client side by the messaging client 104 and on the sever side by the application servers 114. These subsystems include, for example, an ephemeral timer system 202, a collection management system 204, an augmentation system 208, a map system 210, a game system 212, and an external resource system 220.

The ephemeral timer system 202 is responsible for enforcing the temporary or time-limited access to content by the messaging client 104 and the messaging server 118. The ephemeral timer system 202 incorporates a number of timers that, based on duration and display parameters associated with a message, or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the messaging client 104. Further details regarding the operation of the ephemeral timer system 202 are provided below.

The collection management system 204 is responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system 204 may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client 104.

The collection management system 204 also includes a curation interface 206 that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface 206 enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system 204 employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain examples, compensation may be paid to a user for the inclusion of user-generated content into a collection. In such cases, the collection management system 204 operates to automatically make payments to such users for the use of their content.

The augmentation system 208 provides various functions that enable a user to augment (e.g., annotate or otherwise modify or edit) media content associated with a message. For example, the augmentation system 208 provides functions related to the generation and publishing of media overlays for messages processed by the messaging system 100. The augmentation system 208 operatively supplies a media overlay or augmentation (e.g., an image filter) to the messaging client 104 based on a geolocation of the client device 102. In another example, the augmentation system 208 operatively supplies a media overlay to the messaging client 104 based on other information, such as social network information of the user of the client device 102. A media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied

to a media content item (e.g., a photo) at the client device 102. For example, the media overlay may include text, a graphical element, or image that can be overlaid on top of a photograph taken by the client device 102. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the augmentation system 208 uses the geolocation of the client device 102 to identify a media overlay that includes the name of a merchant at the geolocation of the client device 102. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database 126 and accessed through the database server 120.

In some examples, the augmentation system 208 provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The augmentation system 208 generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

In other examples, the augmentation system 208 provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the augmentation system 208 associates the media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time. The augmentation system 208 communicates with the image processing server 122 to obtain AR experiences and presents identifiers of such experiences in one or more user interfaces (e.g., as icons over a real-time image or video or as thumbnails or icons in interfaces dedicated for presented identifiers of AR experiences). Once an AR experience is selected, one or more images, videos, or AR graphical elements are retrieved and presented as an overlay on top of the images or video captured by the client device 102. In some cases, the camera is switched to a front-facing view (e.g., the front-facing camera of the client device 102 is activated in response to activation of a particular AR experience) and the images from the front-facing camera of the client device 102 start being displayed on the client device 102 instead of the rear-facing camera of the client device 102. The one or more images, videos, or AR graphical elements are retrieved and presented as an overlay on top of the images that are captured and displayed by the front-facing camera of the client device 102.

In other examples, the augmentation system 208 is able to communicate and exchange data with another augmentation system 208 on another client device 102 and with the server via the network 112. The data exchanged can include a session identifier that identifies the shared AR session, a transformation between a first client device 102 and a second client device 102 (e.g., a plurality of client devices 102 include the first and second devices) that is used to align the shared AR session to a common point of origin, a common coordinate frame, functions (e.g., commands to invoke functions) as well as other payload data (e.g., text, audio, video or other multimedia data).

The augmentation system 208 sends the transformation to the second client device 102 so that the second client device 102 can adjust the AR coordinate system based on the transformation. In this way, the first and second client devices 102 synch up their coordinate systems and frames for displaying content in the AR session. Specifically, the

augmentation system **208** computes the point of origin of the second client device **102** in the coordinate system of the first client device **102**. The augmentation system **208** can then determine an offset in the coordinate system of the second client device **102** based on the position of the point of origin from the perspective of the second client device **102** in the coordinate system of the second client device **102**. This offset is used to generate the transformation so that the second client device **102** generates AR content according to a common coordinate system or frame as the first client device **102**.

The augmentation system **208** can communicate with the client device **102** to establish individual or shared AR sessions. The augmentation system **208** can also be coupled to the messaging server **118** to establish an electronic group communication session (e.g., group chat, instant messaging) for the client devices **102** in a shared AR session. The electronic group communication session can be associated with a session identifier provided by the client devices **102** to gain access to the electronic group communication session and to the shared AR session. In one example, the client devices **102** first gain access to the electronic group communication session and then obtain the session identifier in the electronic group communication session that allows the client devices **102** to access to the shared AR session. In some examples, the client devices **102** are able to access the shared AR session without aid or communication with the augmentation system **208** in the application servers **114**.

The map system **210** provides various geographic location functions, and supports the presentation of map-based media content and messages by the messaging client **104**. For example, the map system **210** enables the display of user icons or avatars (e.g., stored in profile data **316**) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the messaging system **100** from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the messaging client **104**. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the messaging system **100** via the messaging client **104**, with this location and status information being similarly displayed within the context of a map interface of the messaging client **104** to selected users.

The game system **212** provides various gaming functions within the context of the messaging client **104**. The messaging client **104** provides a game interface providing a list of available games (e.g., web-based games or web-based applications) that can be launched by a user within the context of the messaging client **104**, and played with other users of the messaging system **100**. The messaging system **100** further enables a particular user to invite other users to participate in the play of a specific game, by issuing invitations to such other users from the messaging client **104**. The messaging client **104** also supports both voice and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

The external resource system **220** provides an interface for the messaging client **104** to communicate with external app(s) servers **110** to launch or access external resources. Each external resource (apps) server **110** hosts, for example, a markup language (e.g., HTML5) based application or small-scale version of an external application (e.g., game,

utility, payment, or ride-sharing application that is external to the messaging client **104**). The messaging client **104** may launch a web-based resource (e.g., application) by accessing the HTML5 file from the external resource (apps) servers **110** associated with the web-based resource. In certain examples, applications hosted by external resource servers **110** are programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the messaging server **118**. The SDK includes APIs with functions that can be called or invoked by the web-based application. In certain examples, the messaging server **118** includes a JavaScript library that provides a given third-party resource access to certain user data of the messaging client **104**. HTML5 is used as an example technology for programming games, but applications and resources programmed based on other technologies can be used.

In order to integrate the functions of the SDK into the web-based resource, the SDK is downloaded by an external resource (apps) server **110** from the messaging server **118** or is otherwise received by the external resource (apps) server **110**. Once downloaded or received, the SDK is included as part of the application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the messaging client **104** into the web-based resource.

The SDK stored on the messaging server **118** effectively provides the bridge between an external resource (e.g., third-party or external applications **109** or applets and the messaging client **104**). This provides the user with a seamless experience of communicating with other users on the messaging client **104**, while also preserving the look and feel of the messaging client **104**. To bridge communications between an external resource and a messaging client **104**, in certain examples, the SDK facilitates communication between external resource servers **110** and the messaging client **104**. In certain examples, a Web ViewJavaScript-Bridge running on a client device **102** establishes two one-way communication channels between an external resource and the messaging client **104**. Messages are sent between the external resource and the messaging client **104** via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

By using the SDK, not all information from the messaging client **104** is shared with external resource servers **110**. The SDK limits which information is shared based on the needs of the external resource. In certain examples, each external resource server **110** provides an HTML5 file corresponding to the web-based external resource to the messaging server **118**. The messaging server **118** can add a visual representation (such as a box art or other graphic) of the web-based external resource in the messaging client **104**. Once the user selects the visual representation or instructs the messaging client **104** through a GUI of the messaging client **104** to access features of the web-based external resource, the messaging client **104** obtains the HTML5 file and instantiates the resources necessary to access the features of the web-based external resource.

The messaging client **104** presents a graphical user interface (GUI) (e.g., a landing page or title screen) for an external resource. During, before, or after presenting the landing page or title screen, the messaging client **104** determines whether the launched external resource has been previously authorized to access user data of the messaging client **104**. In response to determining that the launched

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external resource has been previously authorized to access user data of the messaging client **104**, the messaging client **104** presents another GUI of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the messaging client **104**, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the messaging client **104** slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle of or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the messaging client **104** adds the external resource to a list of authorized external resources and allows the external resource to access user data from the messaging client **104**. In some examples, the external resource is authorized by the messaging client **104** to access the user data in accordance with an OAuth 2 framework.

The messaging client **104** controls the type of user data that is shared with external resources based on the type of external resource being authorized. For example, external resources that include full-scale external applications (e.g., a third-party or external application **109**) are provided with access to a first type of user data (e.g., only two-dimensional (2D) avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of external applications (e.g., web-based versions of third-party applications) are provided with access to a second type of user data (e.g., payment information, 2D avatars of users, 3D avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

A synthetic view system **224** accesses a first image depicting a first portion of a body of a person from a first camera view. The synthetic view system **224** detects a fashion item in the first image that is being worn by the person and generates, based on the first image, a second image that depicts a second portion of the body of the person from a second camera view. The synthetic view system **224** generates an augmented reality (AR) item that visually resembles the fashion item from the second camera view and modifies the second image with the AR item to present a synthetic view of the second portion of the body of the person wearing the AR item corresponding to the fashion item. The AR item visually resembles the fashion item by having an identical or similar look as the fashion item. Namely, the AR item has the same visual features and attributes as the fashion item.

An illustrative implementation of the synthetic view system **224** is shown and described in connection with FIG. 5 below.

Data Architecture

FIG. 3 is a schematic diagram illustrating data structures **300**, which may be stored in the database **126** of the messaging server system **108**, according to certain examples. While the content of the database **126** is shown to comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

The database **126** includes message data stored within a message table **302**. This message data includes, for any particular one message, at least message sender data, mes-

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sage recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table **302**, are described below with reference to FIG. 4.

An entity table **306** stores entity data, and is linked (e.g., referentially) to an entity graph **308** and profile data **316**. Entities for which records are maintained within the entity table **306** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the messaging server system **108** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

The entity graph **308** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization) interested-based or activity-based, merely for example.

The profile data **316** stores multiple types of profile data about a particular entity. The profile data **316** may be selectively used and presented to other users of the messaging system **100**, based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data **316** includes, for example, a username, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the messaging system **100**, and on map interfaces displayed by messaging clients **104** to other users. The collection of avatar representations may include "status avatars," which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

Where the entity is a group, the profile data **316** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

The database **126** also stores augmentation data, such as overlays or filters, in an augmentation table **310**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **304**) and images (for which data is stored in an image table **312**).

The database **126** can also store data pertaining to individual and shared AR sessions. This data can include data communicated between an AR session client controller of a first client device **102** and another AR session client controller of a second client device **102**, and data communicated between the AR session client controller and the augmentation system **208**. Data can include data used to establish the common coordinate frame of the shared AR scene, the transformation between the devices, the session identifier, images depicting a body, skeletal joint positions, wrist joint positions, feet, and so forth.

Filters, in one example, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the messaging client **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface

by the messaging client 104, based on geolocation information determined by a Global Positioning System (GPS) unit of the client device 102.

Another type of filter is a data filter, which may be selectively presented to a sending user by the messaging client 104, based on other inputs or information gathered by the client device 102 during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a client device 102, or the current time.

Other augmentation data that may be stored within the image table 312 includes AR content items (e.g., corresponding to applying AR experiences). An AR content item or AR item may be a real-time special effect and sound that may be added to an image or a video.

As described above, augmentation data includes AR content items, overlays, image transformations, AR images, and similar terms that refer to modifications that may be applied to image data (e.g., videos or images). This includes real-time modifications, which modify an image as it is captured using device sensors (e.g., one or multiple cameras) of a client device 102 and then displayed on a screen of the client device 102 with the modifications. This also includes modifications to stored content, such as video clips in a gallery that may be modified. For example, in a client device 102 with access to multiple AR content items, a user can use a single video clip with multiple AR content items to see how the different AR content items will modify the stored clip. For example, multiple AR content items that apply different pseudorandom movement models can be applied to the same content by selecting different AR content items for the content. Similarly, real-time video capture may be used with an illustrated modification to show how video images currently being captured by sensors of a client device 102 would modify the captured data. Such data may simply be displayed on the screen and not stored in memory, or the content captured by the device sensors may be recorded and stored in memory with or without the modifications (or both). In some systems, a preview feature can show how different AR content items will look within different windows in a display at the same time. This can, for example, enable multiple windows with different pseudorandom animations to be viewed on a display at the same time.

Data and various systems using AR content items or other such transform systems to modify content using this data can thus involve detection of objects (e.g., faces, hands, bodies, cats, dogs, surfaces, objects, etc.), tracking of such objects as they leave, enter, and move around the field of view in video frames, and the modification or transformation of such objects as they are tracked. In various examples, different methods for achieving such transformations may be used. Some examples may involve generating a three-dimensional mesh model of the object or objects, and using transformations and animated textures of the model within the video to achieve the transformation. In other examples, tracking of points on an object may be used to place an image or texture (which may be two dimensional or three dimensional) at the tracked position. In still further examples, neural network analysis of video frames may be used to place images, models, or textures in content (e.g., images or frames of video). AR content items thus refer both to the images, models, and textures used to create transformations in content, as well as to additional modeling and analysis information needed to achieve such transformations with object detection, tracking, and placement.

Real-time video processing can be performed with any kind of video data (e.g., video streams, video files, etc.) saved in a memory of a computerized system of any kind. For example, a user can load video files and save them in a memory of a device or can generate a video stream using sensors of the device. Additionally, any objects can be processed using a computer animation model, such as a human's face and parts of a human body, animals, or non-living things such as chairs, cars, or other objects.

In some examples, when a particular modification is selected along with content to be transformed, elements to be transformed are identified by the computing device, and then detected and tracked if they are present in the frames of the video. The elements of the object are modified according to the request for modification, thus transforming the frames of the video stream. Transformation of frames of a video stream can be performed by different methods for different kinds of transformation. For example, for transformations of frames mostly referring to changing forms of an object's elements, characteristic points for each element of an object are calculated (e.g., using an Active Shape Model (ASM) or other known methods). Then, a mesh based on the characteristic points is generated for each of the at least one element of the object. This mesh is used in the following stage of tracking the elements of the object in the video stream. In the process of tracking, the mentioned mesh for each element is aligned with a position of each element. Then, additional points are generated on the mesh. A set of first points is generated for each element based on a request for modification, and a set of second points is generated for each element based on the set of first points and the request for modification. Then, the frames of the video stream can be transformed by modifying the elements of the object on the basis of the sets of first and second points and the mesh. In such a method, a background of the modified object can be changed or distorted as well by tracking and modifying the background.

In some examples, transformations changing some areas of an object using its elements can be performed by calculating characteristic points for each element of an object and generating a mesh based on the calculated characteristic points. Points are generated on the mesh, and then various areas based on the points are generated. The elements of the object are then tracked by aligning the area for each element with a position for each of the at least one element, and properties of the areas can be modified based on the request for modification, thus transforming the frames of the video stream. Depending on the specific request for modification, properties of the mentioned areas can be transformed in different ways. Such modifications may involve changing color of areas; removing at least some part of areas from the frames of the video stream; including one or more new objects into areas which are based on a request for modification; and modifying or distorting the elements of an area or object. In various examples, any combination of such modifications or other similar modifications may be used. For certain models to be animated, some characteristic points can be selected as control points to be used in determining the entire state-space of options for the model animation.

In some examples of a computer animation model to transform image data using face detection, the face is detected on an image with use of a specific face detection algorithm (e.g., Viola-Jones). Then, an ASM algorithm is applied to the face region of an image to detect facial feature reference points.

Other methods and algorithms suitable for face detection can be used. For example, in some examples, features are located using a landmark, which represents a distinguishable point present in most of the images under consideration. For facial landmarks, for example, the location of the left eye pupil may be used. If an initial landmark is not identifiable (e.g., if a person has an eyepatch), secondary landmarks may be used. Such landmark identification procedures may be used for any such objects. In some examples, a set of landmarks forms a shape. Shapes can be represented as vectors using the coordinates of the points in the shape. One shape is aligned to another with a similarity transform (allowing translation, scaling, and rotation) that minimizes the average Euclidean distance between shape points. The mean shape is the mean of the aligned training shapes.

In some examples, a search is started for landmarks from the mean shape aligned to the position and size of the face determined by a global face detector. Such a search then repeats the steps of suggesting a tentative shape by adjusting the locations of shape points by template matching of the image texture around each point and then conforming the tentative shape to a global shape model until convergence occurs. In some systems, individual template matches are unreliable, and the shape model pools the results of the weak template matches to form a stronger overall classifier. The entire search is repeated at each level in an image pyramid, from coarse to fine resolution.

A transformation system can capture an image or video stream on a client device (e.g., the client device **102**) and perform complex image manipulations locally on the client device **102** while maintaining a suitable user experience, computation time, and power consumption. The complex image manipulations may include size and shape changes, emotion transfers (e.g., changing a face from a frown to a smile), state transfers (e.g., aging a subject, reducing apparent age, changing gender), style transfers, graphical element application, and any other suitable image or video manipulation implemented by a convolutional neural network that has been configured to execute efficiently on the client device **102**.

In some examples, a computer animation model to transform image data can be used by a system where a user may capture an image or video stream of the user (e.g., a selfie) using a client device **102** having a neural network operating as part of a messaging client **104** operating on the client device **102**. The transformation system operating within the messaging client **104** determines the presence of a face within the image or video stream and provides modification icons associated with a computer animation model to transform image data, or the computer animation model can be present as associated with an interface described herein. The modification icons include changes that may be the basis for modifying the user's face within the image or video stream as part of the modification operation. Once a modification icon is selected, the transformation system initiates a process to convert the image of the user to reflect the selected modification icon (e.g., generate a smiling face on the user). A modified image or video stream may be presented in a GUI displayed on the client device **102** as soon as the image or video stream is captured and a specified modification is selected. The transformation system may implement a complex convolutional neural network on a portion of the image or video stream to generate and apply the selected modification. That is, the user may capture the image or video stream and be presented with a modified result in real-time or near real-time once a modification icon has been selected. Further, the modification may be persistent while the video

stream is being captured, and the selected modification icon remains toggled. Machine-taught neural networks may be used to enable such modifications.

The GUI, presenting the modification performed by the transformation system, may supply the user with additional interaction options. Such options may be based on the interface used to initiate the content capture and selection of a particular computer animation model (e.g., initiation from a content creator user interface). In various examples, a modification may be persistent after an initial selection of a modification icon. The user may toggle the modification on or off by tapping or otherwise selecting the face being modified by the transformation system and store it for later viewing or browse to other areas of the imaging application. Where multiple faces are modified by the transformation system, the user may toggle the modification on or off globally by tapping or selecting a single face modified and displayed within a graphical user interface. In some examples, individual faces, among a group of multiple faces, may be individually modified, or such modifications may be individually toggled by tapping or selecting the individual face or a series of individual faces displayed within the GUI.

A story table **314** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **306**). A user may create a "personal story" in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the messaging client **104** may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal story.

A collection may also constitute a "live story," which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a "live story" may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the messaging client **104**, to contribute content to a particular live story. The live story may be identified to the user by the messaging client **104**, based on his or her location. The end result is a "live story" told from a community perspective.

A further type of content collection is known as a "location story," which enables a user whose client device **102** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may require a second degree of authentication to verify that the end user belongs to a specific organization or other entity (e.g., is a student on the university campus).

As mentioned above, the video table **304** stores video data that, in one example, is associated with messages for which records are maintained within the message table **302**. Similarly, the image table **312** stores image data associated with messages for which message data is stored in the entity table **306**. The entity table **306** may associate various augmentations from the augmentation table **310** with various images and videos stored in the image table **312** and the video table **304**.

The data structures **300** can also store training data for training one or more machine learning techniques (models) to generate 2D bounding boxes. The training data can include a plurality of training videos and corresponding

ground truth bounding boxes. The images and videos can include a mix of all sorts of real-world objects that can appear in different real-world environments, such as different rooms in a home or household. The one or more machine learning techniques or models can be trained to extract features of a received input image or video and establish a relationship between the extracted features and a 2D bounding box of real-world objects depicted in the image or video. Once trained, the machine learning technique can receive a new image or video and can estimate a 2D bounding box for the newly received image or video.

Data Communications Architecture

FIG. 4 is a schematic diagram illustrating a structure of a message 400, according to some examples, generated by a messaging client 104 for communication to a further messaging client 104 or the messaging server 118. The content of a particular message 400 is used to populate the message table 302 stored within the database 126, accessible by the messaging server 118. Similarly, the content of a message 400 is stored in memory as “in-transit” or “in-flight” data of the client device 102 or the application servers 114. A message 400 is shown to include the following example components:

message identifier 402: a unique identifier that identifies the message 400.

message text payload 404: text, to be generated by a user via a user interface of the client device 102, and that is included in the message 400.

message image payload 406: image data, captured by a camera component of a client device 102 or retrieved from a memory component of a client device 102, and that is included in the message 400. Image data for a sent or received message 400 may be stored in the image table 312.

message video payload 408: video data, captured by a camera component or retrieved from a memory component of the client device 102, and that is included in the message 400. Video data for a sent or received message 400 may be stored in the video table 304.

message audio payload 410: audio data, captured by a microphone or retrieved from a memory component of the client device 102, and that is included in the message 400.

message augmentation data 412: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to message image payload 406, message video payload 408, or message audio payload 410 of the message 400. Augmentation data 412 for a sent or received message 400 may be stored in the augmentation table 310.

message duration parameter 414: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload 406, message video payload 408, message audio payload 410) is to be presented or made accessible to a user via the messaging client 104.

message geolocation parameter 416: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter 416 values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image within the message image payload 406, or a specific video in the message video payload 408).

message story identifier 418: identifier values identifying one or more content collections (e.g., “stories” identi-

fied in the story table 314) with which a particular content item in the message image payload 406 of the message 400 is associated. For example, multiple images within the message image payload 406 may each be associated with multiple content collections using identifier values.

message tag 420: each message 400 may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload 406 depicts an animal (e.g., a lion), a tag value may be included within the message tag 420 that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition.

message sender identifier 422: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device 102 on which the message 400 was generated and from which the message 400 was sent.

message receiver identifier 424: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device 102 to which the message 400 is addressed.

The contents (e.g., values) of the various components of message 400 may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload 406 may be a pointer to (or address of) a location within an image table 312. Similarly, values within the message video payload 408 may point to data stored within a video table 304, values stored within the message augmentation data 412 may point to data stored in an augmentation table 310, values stored within the message story identifier 418 may point to data stored in a story table 314, and values stored within the message sender identifier 422 and the message receiver identifier 424 may point to user records stored within an entity table 306.

Synthetic View System

FIG. 5 is a block diagram showing an example synthetic view system 224, according to example examples. The synthetic view system 224 includes a set of components 510 that operate on a set of input data (e.g., an image 501 depicting a person wearing a real-world object or AR object, such as a fashion item, from a first camera view (e.g., a frontal view of the person)). The synthetic view system 224 includes an image access module 512, a fashion item detection module 514, an AR item generation module 517, a new view image generation module 516, an image modification module 518, and an image display module 520. All or some of the components of the synthetic view system 224 can be implemented by a server, in which case, the image 501 is provided to the server by the client device 102. In some cases, some or all of the components of the synthetic view system 224 can be implemented by the client device 102 or can be distributed across a set of client devices 102.

In some examples, the synthetic view system 224 accesses a first image depicting a first portion of a body of a person from a first camera view. The synthetic view system 224 detects a fashion item in the first image that is being worn by the person depicted in the image, the fashion item being depicted from the first camera view. The synthetic view system 224 generates, based on the first image, a second image that depicts a second portion of the body of the person from a second camera view. The synthetic view system 224 generates, based on the fashion item detected in the first image, an AR item that visually resembles the

fashion item from the second camera view. The synthetic view system 224 modifies the second image with the AR item to present a synthetic view of the second portion of the body of the person wearing the AR item corresponding to the fashion item.

In some examples, the fashion item in the first image includes an AR fashion item. In some examples, the first camera view of the fashion item includes a front view of the AR fashion item. In some examples, the second camera view of the fashion item includes a side view of the AR fashion item. In some examples, the first portion of the body of the person includes a front of a face of the person and the second portion of the body of the person includes a side of the face of the person.

In some examples, the synthetic view system 224 applies a pre-trained generative machine learning model to the first image to generate the second image. In some examples, the synthetic view system 224 applies a 3D rendering process to the first image to generate the second image.

In some examples, the synthetic view system 224 activates a front-facing camera of a device to capture a depiction of the first portion of the body of the person from the first camera view. The synthetic view system 224 synthetically generates the second image to approximate a depiction of the second portion of the body from the second camera view based on the depiction of the first portion of the body of the person from the first camera view.

In some examples, the synthetic view system 224 presents the second image together with the first image. In some examples, the second image is overlaid on top of the first image. In some examples, the second image is presented adjacent to the first image.

In some examples, the synthetic view system 224 synthetically rotates the body of the person to generate the second image that depicts the second portion of the body of the person from the second camera view. In some examples, the fashion item includes an earring, the first portion of the body includes a front view of an ear of the person wearing the earring, and the synthetic view includes a side view of the ear of the person wearing the earring.

In some examples, the fashion item includes a shoe, the first portion of the body includes a front view of a leg and foot of the person wearing the shoe, and the synthetic view includes a side view of the leg and foot of the person wearing the shoe. In some examples, the synthetic view system 224 overlays the AR item on top of the second image to generate the synthetic view.

In some examples, the synthetic view system 224 receives input that selects the second camera view. The second image and the synthetic image can be generated in response to receiving the input. In some examples, the second camera view includes a view of the body of the person from a closer distance to a camera relative to a distance of the body of the person to the camera corresponding to the first camera view.

For example, the image access module 512 can access a real-time camera feed from a front-facing or rear-facing camera of the client device 102. In some other examples, the image access module 512 can receive a camera feed that has been previously recorded or stored, such as from a friend or another user associated with another client device 102. The camera feed can depict a real-world person or a body part of a real-world person from a first camera view (e.g., a front view of the body part of the person). The image access module 512 provides the camera feed that includes one or more frames that depict the person or body part of the person from the first camera view to the fashion item detection module 514.

For example, as shown in FIG. 6, the image access module 512 receives a video 600 that includes a frame 601 that depicts a face of a person 610 from a front view (e.g., a first camera view). The face of the person 610 includes a front view of the ear 620 that is depicted as wearing a fashion item 630, such as an earring. The earring can be a real-world earring or can be an AR earring corresponding to a virtual try-on experience. For example, the client device 102 can receive input that selects or activates a virtual try-on experience by selecting an icon (not shown) corresponding to the virtual try-on experience. In response, the image access module 512 generates the frame 601 in which a virtual fashion item corresponding to the virtual try-on experience is presented as the fashion item 630. In some cases, the frame 601 includes a second fashion item (e.g., another earring depicted as being worn on another ear of the person 610).

In some examples, the image access module 512 can receive input that selects a new view icon (not shown). The new view icon can be presented on top of or adjacent to the frame 601. The new view icon can allow a user to specify a new view (e.g., a particular camera angle or side view) or can present a menu that lists various types of camera views (e.g., top view, side view, view from below, and so forth). In response to receiving the input that selects the new view icon that identifies the new camera view, the image access module 512 provides the video 600 to the fashion item detection module 514 to automate the generation and presentation of a synthesized image or image portion corresponding to the fashion item 630. The synthesized image can depict the person or body part of the person wearing the fashion item 630 from the selected new camera view (e.g., a side view).

Referring back to FIG. 5, the fashion item detection module 514 implements one or more machine learning models (e.g., one or more neural networks) that have been trained to detect and track one or more fashion items in one or more video frames. For example, during training, the machine learning model of the fashion item detection module 514 receives a given training image (or video) from a plurality of training images (or videos) that depict one or more persons wearing fashion items. The plurality of training images are associated with corresponding ground truth markers or indications of the types and locations of the fashion items depicted in the training images and can be received or accessed from training image data stored in data structures 300. The fashion item detection module 514 applies one or more machine learning models to a given training image. The fashion item detection module 514 generates estimated tracking information for a fashion item depicted in the given training image.

The fashion item detection module 514 obtains a known or predetermined ground-truth tracking information corresponding to the given training image. The new view image generation module 516 compares (computes a deviation between) the estimated tracking information with the ground truth tracking information. Based on a difference threshold of the comparison (or deviation), the fashion item detection module 514 updates one or more coefficients or parameters and obtains one or more additional training images from the training data.

After a specified number of epochs or batches of training images have been processed and/or when a difference threshold (or deviation) (computed as a function of a difference or deviation between the estimated tracking information and the ground-truth tracking information) reaches a specified value, the fashion item detection module 514

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completes training, and the parameters and coefficients of the fashion item detection module **514** are stored as a trained machine learning technique.

In some examples, the fashion item detection module **514** can be used to augment a received image or video with a fashion item corresponding to an AR experience by overlaying one or more AR fashion items on the received image or video. The fashion item detection module **514** can track a real-world object of interest (e.g., ears) of a person using one or more previously trained machine learning models and can place the AR object (e.g., AR earrings) relative to the tracked real-world object of interest.

The fashion item detection module **514** can continuously track the fashion item that is depicted in the image or video. The fashion item detection module **514** provides the tracking information for the fashion item to the new view image generation module **516** and to the AR item generation module **517**. The new view image generation module **516** can selectively, dynamically and automatically synthesize a second image that depicts the person in the image accessed by the image access module **512** from a different camera angle or selected camera view.

Specifically, the new view image generation module **516** can receive the image or video from the image access module **512**. The new view image generation module **516** can also receive input that indicates the new camera view that is used to synthesize a new image. For example, the input can request to synthesize a view of the person from a second camera view, such as a side view or perspective.

In some examples, the new view image generation module **516** can apply one or more pre-trained generative machine learning models corresponding to the selected new camera view to the image received from the image access module **512**. The one or more pre-trained generative machine learning models generate a synthesized image and output the synthesized image as a second image that depicts the body part of the person in the original image from the selected new camera view. In some examples, each of the one or more pre-trained generative machine learning models is trained based on respective training data. Particularly, during training, the machine learning model of the new view image generation module **516** receives a given training image (or video) from a plurality of training images (or videos) that depict one or more persons wearing fashion items from a first camera view (e.g., a front view). The plurality of training images is associated with corresponding ground truth images of the same persons wearing the fashion items from a second camera view (e.g., a side view) and can be received or accessed from training image data stored in data structures **300**. The new view image generation module **516** applies one or more machine learning models to a given training image depicting a person wearing a fashion item or not wearing a fashion item from a first camera view perspective (e.g., a front view of the body part of the person, such as the front of the face or front of the legs). In some cases, the new view image generation module **516** applies one or more trained machine learning models to determine the type of view of the body part in the image (e.g., to determine if the current view is a front view or a side view). The new view image generation module **516** generates an estimated image of the person from a second camera view perspective (e.g., a side view of the body part of the person, such as a side of the face or side of the legs).

The new view image generation module **516** obtains a known or predetermined ground-truth image (that has been captured and depicts the same person from the second camera view) corresponding to the given training image.

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The new view image generation module **516** compares (computes a deviation between) the estimated image with the ground truth image. Based on a difference threshold of the comparison (or deviation), the new view image generation module **516** updates one or more coefficients or parameters and obtains one or more additional training images from the training data.

After a specified number of epochs or batches of training images have been processed and/or when a difference threshold (or deviation) (computed as a function of a difference or deviation between the estimated image and the ground-truth tracking image) reaches a specified value, the new view image generation module **516** completes training, and the parameters and coefficients of the new view image generation module **516** are stored as a trained machine learning technique.

In some examples, the new view image generation module **516** can apply a 3D rendering process corresponding to the selected new camera view to the image received from the image access module **512**. The output of the 3D rendering process is a synthesized image that depicts the person depicted in the image received from the image access module **512** from the selected new camera view. In this way (using the generative machine learning models and/or using the 3D rendering process), the new view image generation module **516** synthetically rotates the body part of the person depicted in the image received from the image access module **512** (e.g., captured by the camera) to synthetically generate a new image of the body part from a different camera view.

For example, as shown in FIG. 7, a user interface **700** is presented in which a synthesized image **701** depicts a person **710** and ear **720** of the person **710** from a different camera view than the camera view depicted and used to capture the frame **601**. Specifically, the synthesized image **701** corresponds to a side view of the face of the person **610** depicted in the frame **601** and is a synthesized version or image **701** depicting a side of the face of the person **610** depicted in the frame **601**. The synthesized image **701** resembles a view of the person **710** that is from a side perspective relative to the view captured by the camera of the client device **102** represented by frame **601**.

The AR item generation module **517** receives the new camera view and the tracking information for the fashion item depicted in the image accessed by the image access module **512**. In some examples, the fashion item detection module **514** can segment the fashion item that is detected and only provides the portion of the image received from the image access module **512** corresponding to the fashion item segmentation. The AR item generation module **517** can generate a 3D model of the fashion item depicted in the image received from the image access module **512**. In some cases, the AR item generation module **517** receives the AR fashion item that is depicted in the image received from the image access module **512**. The AR item generation module **517** applies a 3D object model process to rotate, spin, or modify a view of the 3D model of the fashion item to represent a view of the fashion item from the new camera view. The AR item generation module **517** generates an AR item based on the 3D object model that resembles the fashion item from the new camera view. The AR item generation module **517** provides the AR item representing the fashion item from the new camera view to the new view image generation module **516** and/or the image modification module **518**.

The image modification module **518** can receive the synthesized image of the body part of the person from the

new camera view from the new view image generation module **516** and the AR item representing the fashion item from the new camera view from the AR item generation module **517**. The image modification module **518** combines (e.g., projects or overlays) the AR item with the synthesized image to generate a new image or new synthesized image that depicts the body part of the person wearing the fashion item from the new camera view. In some cases, the image modification module **518** can overlay one or more AR objects on the scene generated by the new view image generation module **516**. The image modification module **518** can receive input from the user that moves (repositions) the AR object within the scene and can update the position, orientation, and/or dimensions of the AR object based on the input.

In some examples, as shown in FIG. 7, the image modification module **518** can generate an image **701** in which the synthesized view of the person **710** (e.g., the synthesized or generated side view of the body part of the person, such as the left ear) is combined with the AR item **730** representing the fashion item (e.g., left earring). The synthesized image **701** depicts the synthesized body part of the person **710** wearing the AR item from a different camera view (e.g., a side camera view) than the frame **601** in which the same body part is depicted as wearing the fashion item (e.g., a front camera view). The image display module **520** receives the image from the image modification module **518** and displays the image on a screen of the client device **102**. The image provided by the image display module **520** can be shared with other users on the messaging system **100**. For example, the image provided by the image display module **520** can be included as part of an advertisement or promotion associated with the real-world object depicted in the image.

In some examples, the image display module **520** presents the synthesized image **701** on top of the frame **601**, such as in a picture-in-picture arrangement. In such cases, the synthesized image **701** is presented as a larger image relative to the frame **601**, and the frame **601** is presented in a window on top of the synthesized image **701**. In some examples, the synthesized image **701** is presented as a smaller image relative to the frame **601**, and the synthesized image **701** is presented in a window on top of the frame **601**. In some examples, the synthesized image **701** is presented next to or on top of or below the frame **601**.

In some examples, the synthetic view system **224** presents a user interface **800**, shown in FIG. 8. The user interface **800** includes an image **801** captured by a front-facing camera or rear-facing camera of the client device **102**. The image **801** depicts a side view of legs **810** of a person wearing a fashion item **820**, such as shoes. The synthetic view system **224** can receive input that requests to view the legs **810** and fashion item **820** from a different camera view, such as a front camera view. In response, the synthetic view system **224** presents a user interface **900**, shown in FIG. 9. In the user interface **900**, the synthetic view system **224** generates and presents a synthetic image **901** in which the legs **810** of the person depicted in image **801** are depicted from a front camera view. The synthetic view system **224** also generates a front camera view of the fashion item **820** and overlays that front camera view of the fashion item **820** on the synthetic image **901**. This results in the synthetic image **901** in which the legs **810** depicted as wearing the fashion item **820** from a side camera view are depicted as wearing the fashion item **820** from a front camera view.

FIG. 10 is a flowchart of a process **1000**, in accordance with some examples. Although the flowchart describes the

operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed. A process may correspond to a method, a procedure, and the like. The steps of methods may be performed in whole or in part, may be performed in conjunction with some or all of the steps in other methods, and may be performed by any number of different systems or any portion thereof, such as a processor included in any of the systems.

At operation **1001**, the synthetic view system **224** (e.g., a server or client device **102**) accesses a first image depicting a first portion of a body of a person from a first camera view, as discussed above.

At operation **1002**, the synthetic view system **224** detects a fashion item in the first image that is being worn by the person depicted in the image, the fashion item being depicted from the first camera view, as discussed above. The synthetic view system **224** can also detect a body part of the person depicted in the first image.

At operation **1003**, the synthetic view system **224** generates, based on the first image, a second image that depicts a second portion of the body of the person from a second camera view, as discussed above. For example, the second portion of the body can correspond to a different perspective of the same body part (e.g., the first portion of the body) depicted in the first image, such as a side view versus a front view.

At operation **1004**, the synthetic view system **224** generates, based on the fashion item detected in the first image, an AR item that visually resembles the fashion item from the second camera view, as discussed above.

At operation **1005**, the synthetic view system **224** modifies the second image with the AR item to present a synthetic view of the second portion of the body of the person wearing the AR item corresponding to the fashion item, as discussed above.

Machine Architecture

FIG. 11 is a diagrammatic representation of the machine **1100** within which instructions **1108** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1100** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1108** may cause the machine **1100** to execute any one or more of the methods described herein. The instructions **1108** transform the general, non-programmed machine **1100** into a particular machine **1100** programmed to carry out the described and illustrated functions in the manner described. The machine **1100** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1100** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1100** may comprise, but not be limited to, a server computer, a client computer, a PC, a tablet computer, a laptop computer, a netbook, a STB, a PDA, an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1108**, sequentially or otherwise, that specify actions to be taken by the machine **1100**. Further, while only a single machine **1100** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute

the instructions **1108** to perform any one or more of the methodologies discussed herein. The machine **1100**, for example, may comprise the client device **102** or any one of a number of server devices forming part of the messaging server system **108**. In some examples, the machine **1100** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

The machine **1100** may include processors **1102**, memory **1104**, and input/output (I/O) components **1138**, which may be configured to communicate with each other via a bus **1140**. In an example, the processors **1102** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1106** and a processor **1110** that execute the instructions **1108**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **11** shows multiple processors **1102**, the machine **1100** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

The memory **1104** includes a main memory **1112**, a static memory **1114**, and a storage unit **1116**, all accessible to the processors **1102** via the bus **1140**. The main memory **1104**, the static memory **1114**, and the storage unit **1116** store the instructions **1108** embodying any one or more of the methodologies or functions described herein. The instructions **1108** may also reside, completely or partially, within the main memory **1112**, within the static memory **1114**, within a machine-readable medium within the storage unit **1116**, within at least one of the processors **1102** (e.g., within the processor’s cache memory), or any suitable combination thereof, during execution thereof by the machine **1100**.

The I/O components **1138** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1138** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1138** may include many other components that are not shown in FIG. **11**. In various examples, the I/O components **1138** may include user output components **1124** and user input components **1126**. The user output components **1124** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **1126** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a

touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

In further examples, the I/O components **1138** may include biometric components **1128**, motion components **1130**, environmental components **1132**, or position components **1134**, among a wide array of other components. For example, the biometric components **1128** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **1130** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

The environmental components **1132** include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

With respect to cameras, the client device **102** may have a camera system comprising, for example, front cameras on a front surface of the client device **102** and rear cameras on a rear surface of the client device **102**. The front cameras may, for example, be used to capture still images and video of a user of the client device **102** (e.g., “selfies”), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the client device **102** may also include a 360° camera for capturing 360° photographs and videos.

Further, the camera system of a client device **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the client device **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera, and a depth sensor, for example.

The position components **1134** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

Communication may be implemented using a wide variety of technologies. The I/O components **1138** further include communication components **1136** operable to couple the machine **1100** to a network **1120** or devices **1122** via respective coupling or connections. For example, the communication components **1136** may include a network interface component or another suitable device to interface

with the network 1120. In further examples, the communication components 1136 may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices 1122 may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

Moreover, the communication components 1136 may detect identifiers or include components operable to detect identifiers. For example, the communication components 1136 may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Data-glyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components 1136, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

The various memories (e.g., main memory 1112, static memory 1114, and memory of the processors 1102) and storage unit 1116 may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions 1108), when executed by processors 1102, cause various operations to implement the disclosed examples.

The instructions 1108 may be transmitted or received over the network 1120, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components 1136) and using any one of several well-known transfer protocols (e.g., HTTP). Similarly, the instructions 1108 may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices 1122.

Software Architecture

FIG. 12 is a block diagram 1200 illustrating a software architecture 1204, which can be installed on any one or more of the devices described herein. The software architecture 1204 is supported by hardware such as a machine 1202 that includes processors 1220, memory 1226, and I/O components 1238. In this example, the software architecture 1204 can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture 1204 includes layers such as an operating system 1212, libraries 1210, frameworks 1208, and applications 1206. Operationally, the applications 1206 invoke API calls 1250 through the software stack and receive messages 1252 in response to the API calls 1250.

The operating system 1212 manages hardware resources and provides common services. The operating system 1212 includes, for example, a kernel 1214, services 1216, and drivers 1222. The kernel 1214 acts as an abstraction layer between the hardware and the other software layers. For example, the kernel 1214 provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionality. The services 1216 can provide other common services for the other software layers. The drivers 1222 are

responsible for controlling or interfacing with the underlying hardware. For instance, the drivers 1222 can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

The libraries 1210 provide a common low-level infrastructure used by the applications 1206. The libraries 1210 can include system libraries 1218 (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries 1210 can include API libraries 1224 such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in 2D and 3D in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries 1210 can also include a wide variety of other libraries 1228 to provide many other APIs to the applications 1206.

The frameworks 1208 provide a common high-level infrastructure that is used by the applications 1206. For example, the frameworks 1208 provide various GUI functions, high-level resource management, and high-level location services. The frameworks 1208 can provide a broad spectrum of other APIs that can be used by the applications 1206, some of which may be specific to a particular operating system or platform.

In an example, the applications 1206 may include a home application 1236, a contacts application 1230, a browser application 1232, a book reader application 1234, a location application 1242, a media application 1244, a messaging application 1246, a game application 1248, and a broad assortment of other applications such as an external application 1240. The applications 1206 are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications 1206, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the external application 1240 (e.g., an application developed using the ANDROID™ or IOS™ SDK by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the external application 1240 can invoke the API calls 1250 provided by the operating system 1212 to facilitate functionality described herein.

Glossary

“Carrier signal” refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

“Client device” refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, PDA, smartphone, tablet, ultrabook, net-
book, laptop, multi-processor system, microprocessor-based or programmable consumer electronics, game console, STB, or any other communication device that a user may use to access a network.

“Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

“Component” refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions.

Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein.

A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an ASIC. A hardware component may

also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein.

Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time.

Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or

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more processors 1102 or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations 5 may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the 10 processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). 15 In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

“Computer-readable storage medium” refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium,” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure. 20

“Ephemeral message” refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video, and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default 25 setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

“Machine storage medium” refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store 30 executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable 35 read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage medium,” “device-storage medium,” and “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.” 40

“Non-transitory computer-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine. 45

“Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated 50 data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its

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characteristics set or changed in such a manner as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

Changes and modifications may be made to the disclosed examples without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.

What is claimed is:

1. A method comprising:

accessing, by one or more processors of a device, a first image depicting a first portion of a body of a person from a first camera view;

detecting a fashion item in the first image that is being worn by the person in the first image, the fashion item being depicted in the first image from the first camera view;

generating, based on the first image, a second image that depicts a second portion of the body of the person from a second camera view;

generating, based on the fashion item detected in the first image, an augmented reality (AR) item that visually resembles the fashion item from the second camera view; and

modifying the second image with the AR item to present a synthetic view of the second portion of the body of the person wearing the AR item corresponding to the fashion item.

2. The method of claim 1, wherein the fashion item in the first image comprises an AR fashion item.

3. The method of claim 1, wherein the first camera view of the fashion item comprises a front view of the AR fashion item.

4. The method of claim 3, wherein the second camera view of the fashion item comprises a side view of the AR fashion item.

5. The method of claim 1, wherein the first portion of the body of the person comprises a front of a face of the person, and wherein the second portion of the body of the person comprises a side of the face of the person.

6. The method of claim 1, further comprising:

applying a generative machine learning model to the first image to generate the second image.

7. The method of claim 1, further comprising:

applying a three-dimensional (3D) rendering process to the first image to generate the second image.

8. The method of claim 1, further comprising:

activating a front-facing camera of a device to capture a depiction of the first portion of the body of the person from the first camera view; and

synthetically generating the second image to approximate a depiction of the second portion of the body from the second camera view based on the depiction of the first portion of the body of the person from the first camera view.

9. The method of claim 1, further comprising:

presenting the second image together with the first image.

10. The method of claim 9, wherein the second image is overlaid on top of the first image.

11. The method of claim 9, wherein the second image is presented adjacent to the first image.

12. The method of claim 1, further comprising:

synthetically rotating the body of the person to generate the second image that depicts the second portion of the body of the person from the second camera view.

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13. The method of claim 1, wherein the fashion item comprises an earring, wherein the first portion of the body comprises a front view of an ear of the person wearing the earring, and wherein the synthetic view comprises a side view of the ear of the person wearing the earring.

14. The method of claim 1, wherein the fashion item comprises a shoe, wherein the first portion of the body comprises a front view of a leg and foot of the person wearing the shoe, and wherein the synthetic view comprises a side view of the leg and foot of the person wearing the shoe.

15. The method of claim 1, further comprising:
overlaying the AR item on top of the second image to generate the synthetic view.

16. The method of claim 1, further comprising:
receiving input that selects the second camera view, wherein the second image and the synthetic image are generated in response to receiving the input.

17. The method of claim 1, wherein the second camera view comprises a view of the body of the person from a closer distance to a camera relative to a distance of the body of the person to the camera corresponding to the first camera view.

18. A system comprising:

a processor of a device configured to perform operations comprising:

accessing a first image depicting a first portion of a body of a person from a first camera view;

detecting a fashion item in the first image that is being worn by the person in the first image, the fashion item being depicted in the first image from the first camera view;

generating, based on the first image, a second image that depicts a second portion of the body of the person from a second camera view;

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generating, based on the fashion item detected in the first image, an augmented reality (AR) item that visually resembles the fashion item from the second camera view; and

modifying the second image with the AR item to present a synthetic view of the second portion of the body of the person wearing the AR item corresponding to the fashion item.

19. The system of claim 18, wherein the fashion item in the first image comprises an AR fashion item.

20. A non-transitory machine-readable storage medium that includes instructions that, when executed by one or more processors of a device, cause the device to perform operations comprising:

accessing a first image depicting a first portion of a body of a person from a first camera view;

detecting a fashion item in the first image that is being worn by the person in the first image, the fashion item being depicted in the first image from the first camera view;

generating, based on the first image, a second image that depicts a second portion of the body of the person from a second camera view;

generating, based on the fashion item detected in the first image, an augmented reality (AR) item that visually resembles the fashion item from the second camera view; and

modifying the second image with the AR item to present a synthetic view of the second portion of the body of the person wearing the AR item corresponding to the fashion item.

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